Question 1:
What is the expected output of the following code?

```
print(type(1 / 2))
```

A. <type 'int' >
B. <type 'float' >
C. <type 'tuple' >
D. <type 'number' >
E. <type 'double' >

Question 2:
You are creating a function that reads a data file and prints each line of the file. You write the following code.

```
1  import os                        # Line 01
2  def read_file(file):             # Line 02
3      line = None                  # Line 03
4      if os.path.isfile(file):     # Line 04
5          data = open(file, 'r')   # Line 05
6      while line != '':            # Line 06
7          line = data.readline()   # Line 07
8          print(line)              # Line 08
```

The code attempts to read the file even if the file does not exist. You need to correct the code. Which lines have indentation problems? Each correct answer presents part of the solution. Choose three.

A. Line 07
B. Line 04
C. Line 08
D. Line 02
E. Line 03
F. Line 06
G. Line 05
H. Line 01

Question 3:
Consider the following program to calculate a discount percentage:

```
day = input('Enter the day of the week:')
discount = 3

if day == 'Wednesday':
    discount += 5
elif day == 'Thursday':
    discount += 7
elif day == 'Saturday':
    discount += 10
elif day == 'Sunday':
    discount += 20
else:
    discount += 2
```

Which of the following inputs will get the user a discount of 5 %?

A. Sunday
B. Friday
C. Wednesday
D. Saturday
E. Thursday

Question 4:
What is the expected output of the following code?

```
try:
    if '1' != 1:
        raise FirstError
    else:
        print('FirstError has not occured.')
except FirstError:
    print('FirstError has occured.')
else:
    print('None of the above.')
```

A. The code is erroneous.
B. FirstError has occurred.
C. FirstError has not occurred.
D. None of the above.

Question 5:
What is the expected output of the following code?

```
fruits1 = ['Apple', 'Pear', 'Banana']
fruits2 = fruits1
fruits3 = fruits1[:]

fruits2[0] = 'Cherry'
fruits3[1] = 'Orange'

res = 0

for i in (fruits1, fruits2, fruits3):
    if i[0] == 'Cherry':
        res += 1
    if i[1] == 'Orange':
        res += 10

print(res)
```

A. 11
B. 22
C. 0
D. 12

Question 6:
How many stars will the following snippet print to the monitor?

```
x = 16
while x > 0:
    print('*')
    x //= 2
```

A. The code will enter an infinite loop.
B. three
C. five
D. one

Question 7:
What is the expected output of the following code?

```
data = [(0, 1), (1, 2), (2, 3)]
res = sum(i for j, i in data)
print(res)
```

A. The code is erroneous.
B. 9
C. 3
D. 6

Question 8:
What is the expected output of the following code?

```
def test(x, y=23, z=10):
    print('x is', x, 'and y is', y, 'and z is', z)

test(3, 7)
test(42, z=24)
test(z=60, x=100)
```

A.	1 | x is 3 and y is 7 and z is 10 2 | x is 42 and y is 23 and z is 24 3 | x is 100 and y is 23 and z is 60
B.	1 | x is 3 and y is 7 and z is 10 2 | x is 23 and y is 42 and z is 24 3 | x is 60 and y is 100 and z is 23
C.	The code is erroneous.
D.	1 | x is 7 and y is 3 and z is 10 2 | x is 42 and y is 23 and z is 24 3 | x is 23 and y is 100 and z is 60

Question 9:
What is the expected output of the following code?

```
from random import randint

for i in range(2):
    print(randint(1, 2), end='')
```

A. There are millions of possible combinations, and the exact output cannot be predicted.
B. 11,12, 21, or 22
C. 12 or 21
D. 12

Question 10:
Which of the following open modes allow you to perform read operations? Choose two.

A. r
B. w
C. a
D. r+

Question 11:
The ABC company is creating a program that allows customers to log the number of miles biked. The program will send messages based on how many miles the customer logs. You create the following Python code.

```
???
    name = input('What is your name? ')
    return name

???
    calories = miles * calories_per_mile
    return calories

distance = int(input('How many miles did you bike this week? '))
burn_rate = 50
biker = get_name()
calories_burned = calc_calories(distance, burn_rate)
print(biker + ', you burned about', calories_burned, 'calories.')
```

What would you insert instead of ??? and ???

A. def get_name (biker) :
B. def calc_calories ( ) :
C. def get_name ( ):
D. def calc_calories (miles, burn_rate) :
E. def calc_calories (miles, calories_per_mile):
F. def get_name (name):

Question 12:
Consider the following python code:

```
x1 = '23'
y1 = 7
z1 = x1 * y1

x2 = 42
y2 = 7
z2 = x2 / y2

x3 = 4.7
y3 = 1
z3 = x3 / y3
```

What are the data types of the variables z1, z2 and z3?

A. str , int, float
B. str, str, str
C. str, int, int
D. str, float, float

Question 13:
What is the expected output of the following snippet?

```
s = 'python'
for i in range(len(s)):
    i = s[i].upper()
print(s, end='')
```

A. The code is erroneous.
B. Python
C. PYTHON
D. python
E. P y t h o n
F. P Y T H O N

Question 14:
Which of the following code snippets will print True to the monitor?

A.	1 | x = ‘Peter Wellert’ 2 | y = “Peter Wellert’ .lower( ) 3 | print (x is y)
B.	1 | print (‘t’ in ‘Peter’)
C.	1 | print (‘is’ in ‘This IS Python code.’)
D.	1 | x = [‘Peter’ , ‘Paul’ , ‘Mary’] 2 | y = [‘Peter’ , ‘Paul’ , ‘Mary’] 3 | print (x is y)
E.	1 | x = 42 2 | y = 42 3 | print (x is not y)

Question 15:
By which variable of the sys module can we access command line arguments?

A. argv
B. argsv
C. arguments
D. args

Question 16:
What is the expected output of the following code?

```
data = (1, 2, 4, 8)
data = data[1:-1]
data = data[0]
print(data)
```

A. 2
B. (2)
C. The code is erroneous.
D. (2,)

Question 17:
You want to write a new player to a file and print the whole file content to the monitor. What snippet would you insert in the line indicated below:

```
customer = 'Peter Wellert'
with open('customers.txt', 'a+') as f:
    f.write(customer)
    # insert your code here
    data = f.read()
    print(data)
```

A. f.begin( )
B. f.close( )
C. f.flush( )
D. f.seek( )

Question 18:
What is the expected output of the following code?

```
x = 4.5
y = 2
print(x // y)
```

A. 2
B. 2.0
C. 2.25
D. 2.5

Question 19:

What is the expected output of the following code?

```
num = 2 + 3 * 5
print(Num)
```

A. The code is erroneous.
B. 17.0
C. 17
D. 25

Question 20:
What is the expected output of the following code?

```
nums = [1, 2, 3]
data = ('Peter',) * (len(nums) - nums[::-1][0])
print(data)
```

A. ( )
B. ('Peter')
C. ('Peter' , 'Peter')
D. The code is erroneous.
E. PeterPeter

Question 21:
When an exception occurs, we say that it has been:

A. dropped
B. thrown
C. raised
D. None of the above.

Question 22:
What is the expected output of the following code?

```
data = [1, {}, (2,), (), {3}, [4, 5]]
points = 0

for i in range(len(data)):
    if type(data[i]) == list:
        points += 1
    elif type(data[i]) == tuple:
        points += 10
    elif type(data[i]) == set:
        points += 100
    elif type(data[i]) == dict:
        points += 1000
    else:
        points += 10000

print(points)
```

A. 11112
B. 10212
C. 10221
D. 21102
E. 11121

Question 23:
Which of the following function definitions does not return any value?

A. A function that converts an uppercase letter to lowercase.
B. A function that returns a random integer from I to loo.
C. A function that prints integers from I to 100.
D. None of the above.

Question 24:
What is the expected output of the following code?

```
w = [7, 3, 23, 42]
x = w[1:]
y = w[1:]
z = w
y[0] = 10
z[1] = 20
print(w)
```

A. [7, 3, 23, 42]

B. [7, 20, 23, 42]
C. [10, 20, 23, 42]
D. [10, 20, 42]

Question 25:
What is the expected output of the following code if the user enters 2 and 4?

```
x = float(input())
y = float(input())
print(y ** (1 / x))
```

A. 1.0
B. 0.0
C. 2.0
D. 4.0

Question 26:
You develop a Python application for your company. You want to add notes to your code so other team members will understand it. What should you do?

A. Place the notes inside of parentheses on any line.
B. Place the notes after the last line of code separated by a blank line.
C. Place the notes before the first line of code separated by a blank line.
D. Place the notes after the # sign on any line.

Question 27:
What is the expected output of the following code?

```
try:
    file = open('data.txt', 'r')
    file.write('Hello file!')
except:
    print('An error occurred.')
else:
    print('The content is written successfully.')
```

A. Hello file!
B. The content is written successfully.
C. The code is erroneous.
D. An error occurred.

Question 28:
What is the expected output of the following code?

```
def func(item):
    item += [1]

data = [1, 2, 3, 4]
func(data)
print(len(data))
```

A. 2
B. 5
C. The code is erroneous.
D. 4

Question 29:
What is the expected output of the following code?

```
def test(x=1, y=2):
    x = x + y
    y += 1
    print(x, y)

test()
```

A. 1 1
B. The code is erroneous.
C. 1 3
D. 3 3
E. 3 1

Question 30:
UNICODE is a standard:

A. for coding floating-point numbers.
B. used by coders from universities.
C. like ASCII, but much more expansive.
D. honoured by the whole universe.

Question 31:
You want to print each name of the list on a new line.

```
data = ['Peter', 'Paul', 'Mary', 'Jane']
```

Which statement will you use?

A. print ('\n' .join(data))
B. print (data.concatenate('\n'))
C. print (data.join('\n'))
D. print (data.join ('%\n' , names))

Question 32:
The folder created by Python used to store pyc files is named:

A. __cache__
B. __pycfiles__
C. __pycache__
D. __pyc__

Question 33:
What is the expected output of the following code?

```
print(len([i for i in range(0, -2)]))
```

A. 1
B. 0
C. 2
D. 3

Question 34:
What is the expected output of the following code?

```
def func(x):
    global y
    y = x * x
    return y

func(2)
print(y)
```

A. None
B. The code is erroneous.
C. 2

D. 4

Question 35:
What is the expected output of the following code?

```
def func():
    try:
        print('Monday')
    finally:
        print('Friday')

func()
```

A.	The code is erroneous.
B.	Monday
C.	Friday
	Monday
D.	Friday
E.	Monday Friday
F.	None of these.

Question 36:
How many elements will the following list contain?

```
data = [i for i in range(-1, 2)]
```

A. four
B. two
C. one
D. zero
E. three

Question 37:
Which of the following statements are true regarding the opening modes of a file? Choose three.

A. When you open a file for writing, if the file does not exist, a new file is created.
B. When you open a file for reading, if the file does not exist, an error occurs.
C. When you open a file for writing, if the file exists, the existing file is overwritten with the new file.
D. When you open a file for writing, if the file does not exist, an error occurs.
E. When you open a file for reading, if the file does not exist, the program will open an empty file.

Question 38:
An alternative name for a data structure called a stack is:

A. LIFO
B. FOLO
C. FIFO
D. None of the above.

Question 39:
You want to find the hypotenuse of a right-angled triangle where perpendicular and base are known:

```
# Line 1
parendicular = 3
base = 4
print(hypot(parendicular, base))
```

What is missing in Line 1?

A. from maths import hypot
B. Nothing at all.
C. import maths
D. import hypot

Question 40:
What is the expected output of the following code?

```
data = {'a': 1, 'b': 2, 'c': 3}
print(data['a', 'b'])
```

A. The code is erroneous.
B. (1,2)
C. {'a' :1, 'b' :2}
D. [1, 2]

Question 41:
You have the following file.
index.py:

```
from sys import argv
print(argv[0])
```

You run the file by executing the following command in the terminal.

```
python index.py Hello
```

What is the expected output?

A. Hello
B. IndexError
C. ImportError
D. index.py

Question 42:
What value will be assigned to the x variable?

```
z = 2
y = 1
x = y < z or z > y and y > z or z < y
```

A. False
B. True
C. 0
D. 1

Question 43:

What is the expected output of the following code?

```
x = True
y = False
z = False

if x or y and z:
    print('TRUE')
else:
    print('FALSE')
```

A. The code is erroneous.
B. FALSE
C. TRUE
D. None of the above.

Question 44:
Assuming that all three files x.py, y.py, and z.py reside in the same folder, what will be the output produced by running the z.py file?

x.py:

```
print('x', end='')
```

y.py:

```
import x
print('y', end='')
```

z.py:

```
print('z', end='')
import x
import y
```

A. zyx
B. zxxy
C. The code is erroneous.
D. zxy

Question 45:
Which of the following commands is used to open a file in binary format for writing only?

A. open ('data.txt' , 'wt')
B. open ('data.txt' , 'w')
C. open ('data.txt' , 'wb')
D. open ('data.txt' , 'wb+')

Question 46:
What is the expected output of the following code?

```
def func(x, y=2):
    num = 1
    for i in range(y):
        num = num * x
    return num

print(func(4))
print(func(4, 4))
```

A.	1 | 16 2 | 256
B.	1 | 128 2 | 512
C.	1 | 8 2 | 16
D.	1 | 32 2 | 1024

Question 47:
Select the true statements. Choose two.

A. Different except: branches may have unique exception names.
B. Different except: branches must have unique exception names.
C. You cannot use multiple except: branches to handle multiple exceptions coming from the same try: branch.
D. You can use multiple exceptions: branches to handle multiple exceptions coming from the same try: branch.

Question 48:
What is the expected output of the following code?

```
data = (1,) * 3
data[0] = 2
print(data)
```

A. (2, 1, 1)
B. (1, 1, 1)
C. (2, 2, 2)
D. The code is erroneous.

Question 49:
What is the expected output of the following code?

```
import math
print(dir(math))
```

A. The code is erroneous.
B. A list of all the entities residing in the maths module.
C. The number of all the entities residing in the maths module.
D. A string containing the fully qualified name of the module.

Question 50:
How many stars will the following snippet print to the monitor?

```
data = [[x for x in range(y)] for y in range(3)]

for d in data:
    if len(d) < 2:
        print('*')
```

A. two
B. one
C. three
D. zero

Question 51:
What is the expected output of the following code?

```
print(ord('c') - ord('a'))
```

A. 0
B. 3
C. 2
D. 1

Question 52:
What is the expected output of the following code?

```
print(type(1J))
```

A. <type 'dict' >

B. <type 'float' >
C. <type 'complex' >
D. <type 'unicode' >

Question 53:
What is the expected output of the following code?

```
data = ()
print(data.__len__())
```

A. None
B. The code is erroneous.
C. 1
D. 0

Question 54:
The ** operator ...

A. performs duplicated multiplication.
B. does not exist.
C. performs floating-point multiplication.
D. performs exponentiation.

Question 55:
What is the expected output of the following code?

```
my_list = [[3-i for i in range(3)] for j in range(3)]
result = 0

for i in range(3):
    result += my_list[i][i]

print(result)
```

A. 6
B. 7
C. 2

D. 4

Question 56:
The sys.stderr stream is normally associated with:

A. the keyboard
B. the printer
C. the screen
D. a null device

Question 57:
You are an intern for ABC electric cars company. You must create a function that calculates the average velocity of their vehicles on a 1320 foot (1/4 mile) track. Consider the following code.

```
distance = ???(input('Enter the distance travelled in feet'))
distance_miles = distance/5280  # convert to miles

time = ???(input('Enter the time elapsed in seconds'))
time_hours = time/3600  # convert to hours

velocity = distance_miles/time_hours
print('The average Velocity : ', velocity, 'miles/hour')
```

The output must be as precise as possible. What would you insert instead of ??? and ???

A.	1 | float 2 | float
B.	1 | float 2 | int
C.	1 | int 2 | int
D.	1 | int 2 | float